Missing Blue

Reconstruct the Unknown. Verify What Matters.

AI-Powered Water Infrastructure Intelligence Platform

Jordan | Smart Society & Public Services



Blue

The Jordanian Water Challenge

41%

non-revenue water

National level, Aug 2026

25%

target by 2040

National Water Strategy

Incomplete GIS records

Undocumented segments slow maintenance

The easy losses are being fixed. What remains is loss no one can locate.

Sources: Ministry of Water & Irrigation (Aug 2026); National Water Strategy

Why Current Investigation Takes Too Long

Incomplete network records

Unknown connections

Crews may not know how segments relate

Fragmented evidence

GIS, field notes, and observations are disconnected

Slow field verification

Time is lost before the right site is inspected



1. **Anomaly / Pipe break**



2. **Uncertainty**
   Incomplete information and unknown connections



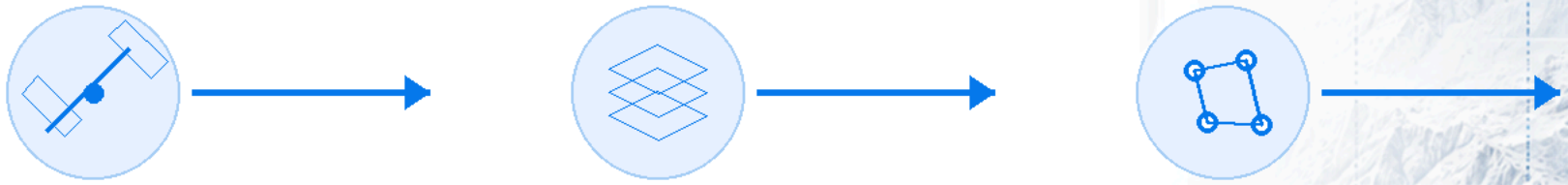
3. **Repeatted investigation**
   Multiple data sources and site checks



4. **Delay**
   More time and cost before resolution

The core problem is not just missing pipes on a map — it is **missing decision-ready knowledge.**

How Missing Blue Works



01 Observe

NISAR L-band InSAR, public pipe maps, terrain, rainfall

02 Simulate

WNTR + EPANET digital twin — Karak: 850 junctions, 63.6 km

03 Infer

ML ranking + Bayesian update over network hypotheses

04 Act

Next-best measurement + work order

A physics-guided AI workflow that combines public data, hydraulic simulation, and explainable decision support.

Why Now

Launched July 30, 2025

NASA–ISRO NISAR satellite

Every 12 days

Repeat L-band radar coverage

Free & open

100,000+ L-band products already public

Free, repeat L-band radar over Jordan didn't exist two years ago.

Missing Blue

What We Built — And What We Do Not Claim

Built Today

- Real NISAR radar processing
- Real public pipe-map matching
- Hydraulic digital twin
- ML inference and Bayesian updates
- Arabic / English work-order workflow

Not Yet Proven

- Confirmed underground leaks
- Field-validated savings
- Paid utility deployment
- Full network truth in Jordan
- Automatic leak confirmation from space

We distinguish observed evidence, simulated behavior, model hypotheses, and field-verified facts.

Missing Blue

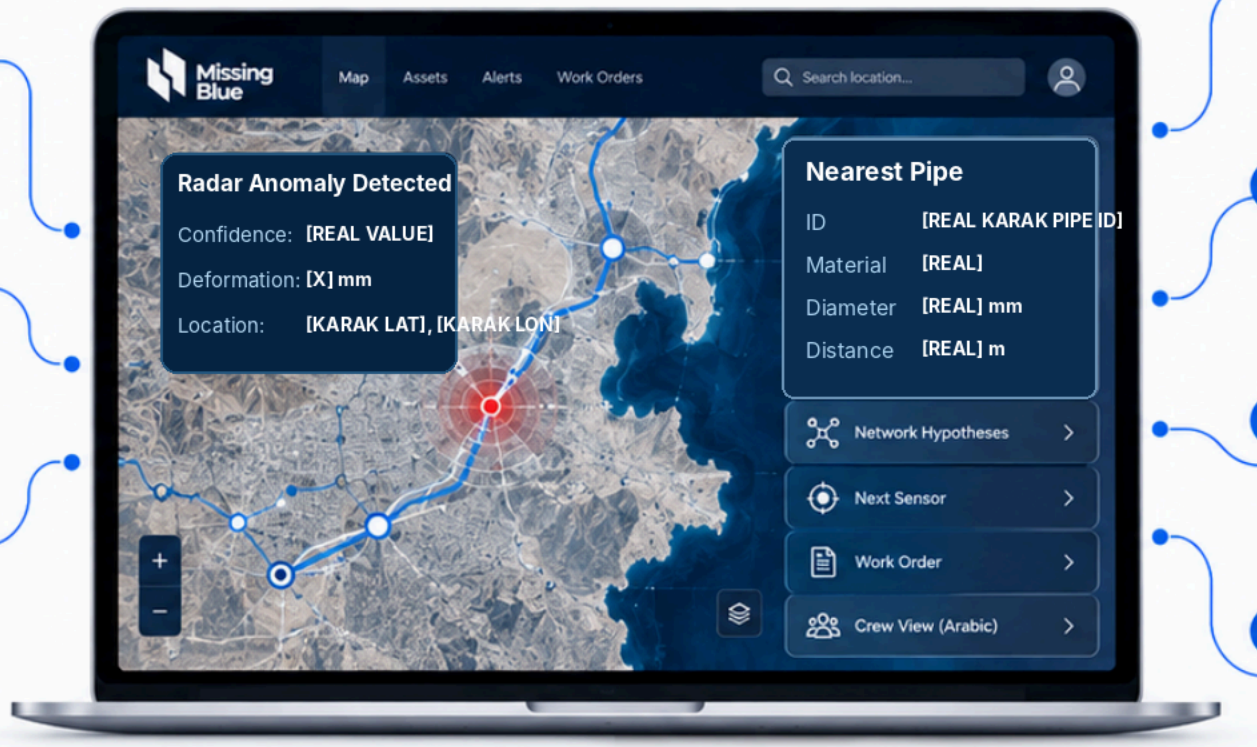
Live Demo

From detection to decision — in one integrated workflow.

1 Detect surface anomaly
NISAR InSAR flags ground deformation or moisture change near the network.

2 Match nearest pipe
System finds the closest asset in the network and retrieves details.

3 Compare network hypotheses
AI evaluates explanations using the hydraulic twin, terrain and history.



4 Recommend next sensor
System suggests the best location for the next field measurement.

5 Generate work order
Ready-to-execute work order with location, risk and recommended steps.

6 Crew reviews in Arabic
Field crew sees simplified guidance in Arabic on mobile.

7 Engineer decides
Engineer reviews AI insights and approves the action.

In our Karak twin, one recommended measurement narrows [N] candidate segments to [M].

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Arabic Crew Workflow

Arabic-first
Native, intuitive interface for field crews.

QR accessible
Quick access to assets on site.

Designed for field crews
Simple, clear, and actionable in the field.

Human review remains required before any engineering decision.



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Pilot and Business Model

Who Pays?

- Water utilities
- WAJ-coordinated programs
- Potential technical collaboration with RJGC

Pilot



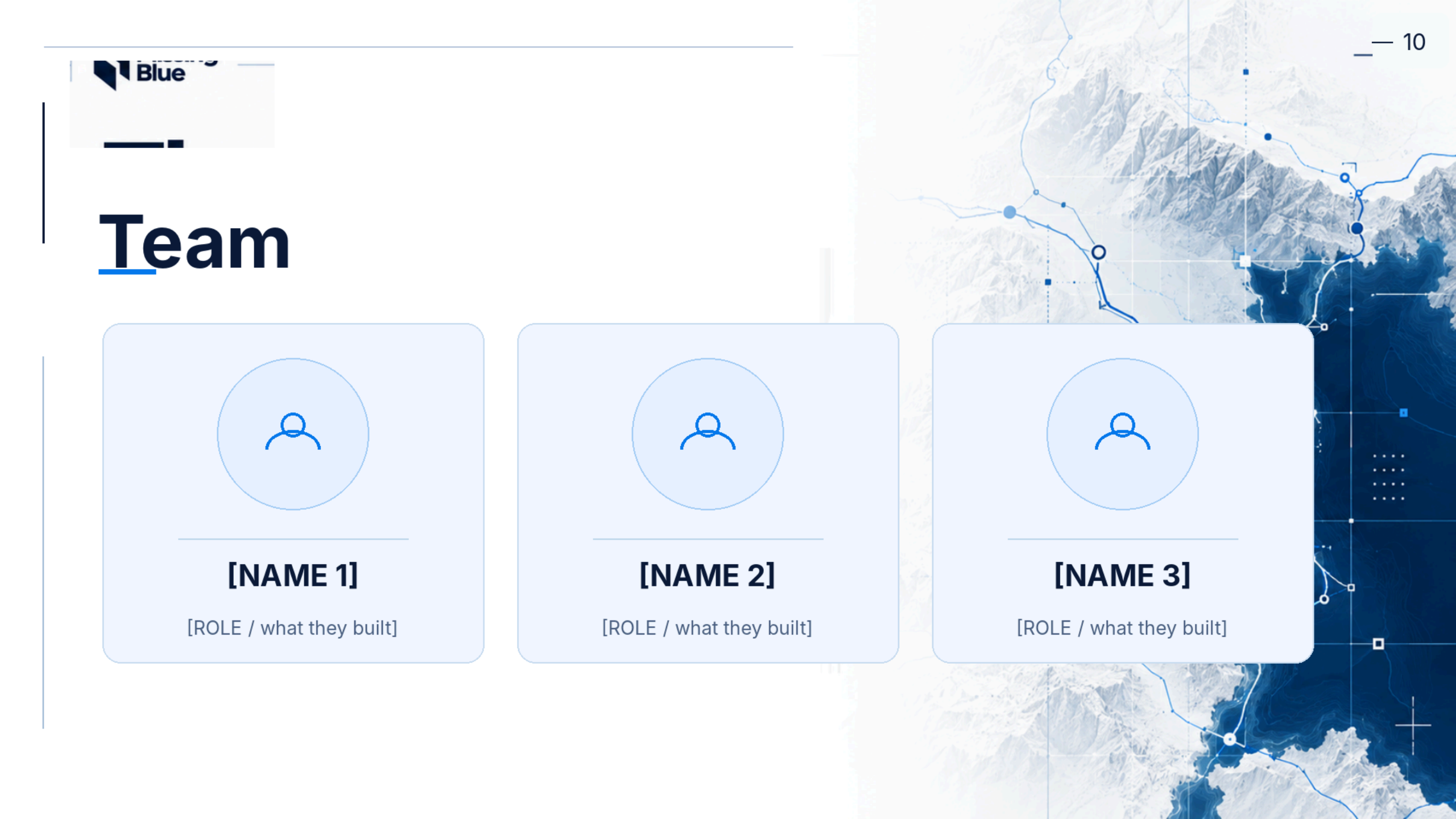
Business Model

- Institutional software
- Implementation and integration
- Training and support

Pilot ask: 90 days, one Karak zone. Utility provides pressure/flow logs + one field crew. Success metric: hit rate of our top-3 recommendations in field checks.

Blue

Team



[NAME 1]

[ROLE / what they built]

[NAME 2]

[ROLE / what they built]

[NAME 3]

[ROLE / what they built]

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Reconstruct the Unknown. Verify What Matters.

We're asking for a 90-day supervised pilot in Karak.

